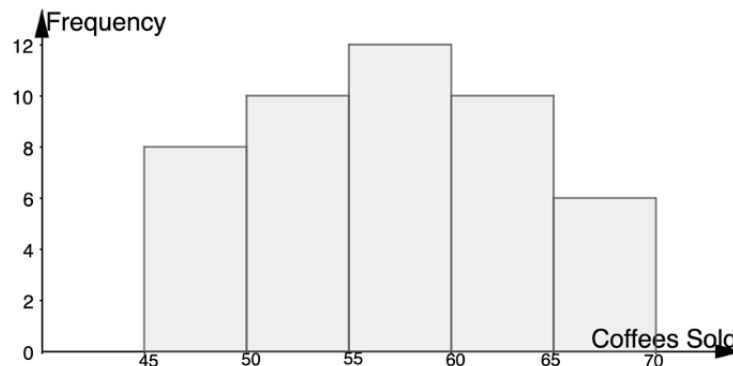


The histogram shows the daily number of coffees sold in a cafe.



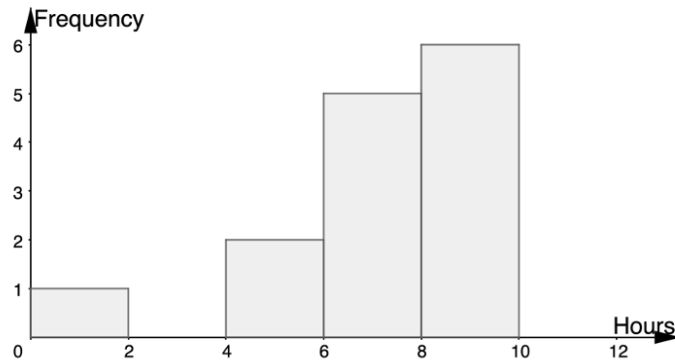
1. Describe the shape of the histogram.
The graph is mostly symmetrical.
2. Explain which measure of center, mean or median, best represents this data.
Mean because the average represents the typical coffees sold which is easily seen in the middle interval because this graph is symmetrical.
3. Explain which measure of variability, range or IQR, best represents this data.
Range because there aren't any outliers.

The numbers of weekly miles run by a marathon runner in training are shown.

15, 15, 15, 16, 17, 17, 17, 18, 18, 19, 20, 21

4. Explain which measure of center, mean or median, best represents this data.
Mean because the data doesn't have outliers.
5. Explain which measure of variability, range or IQR, best represents this data.
Range because there aren't any outliers.

The histogram shows the number of hours of sleep a person got over the past *two* weeks.



6. Describe the shape of the histogram.
Left skewed
7. Explain which measure of center, mean or median, best represents this data.
Median because the graph is skewed, and the mean is potentially affected by the outlier in the first interval.
8. Explain which measure of variability, range or IQR, best represents this data.
IQR because the outlier might cause a misinterpretation of the data.

The ages of the people at a party are shown.

5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 18, 35

9. Explain which measure of center, mean or median, best represents this data.
Median because the outlier, 35, would have an effect on the mean.
10. Explain which measure of variability, range or IQR, best represents this data.
IQR because the outlier would affect the range.